IMEN574 / IDSC574 · PROGRAMMING FOR AX

11

Clustering

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Where We Are

01 Programming with AI

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 Deep Learning and Foundation Models

Lectures 14–22

Outline: Clustering

01 **Similarity and Prototypes**

Choose a geometry

02 **Distributions and Density**

Represent flexible groups

03 **Structure and Validation**

Test the grouping

01 · Similarity and Prototypes

01 **Similarity and Prototypes**
Choose a geometry

02 **Distributions and Density**
Represent flexible groups

03 **Structure and Validation**
Test the grouping

Clustering Questions

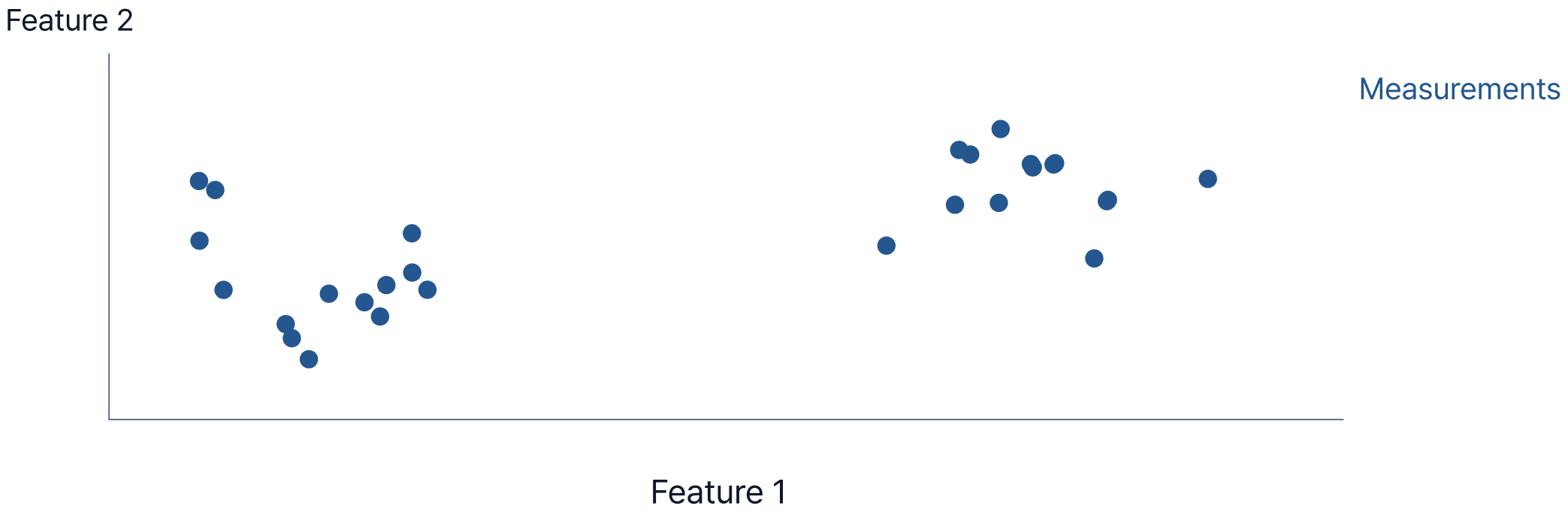
Similarity defines which grouping is useful

Question	Required Choice
What is one object?	Row, person, machine
What is close?	Features, scaling, distance
What is useful?	Compact groups, density, hierarchy

The same rows can support different grouping questions

Unlabeled Observations

The input contains measurements without a target label



These measurements contain two visible concentrations

Distance Geometry

A distance combines feature differences

	A	B	C
A	0	3	4
B	3	0	5
C	4	5	0

$$d(x,z) = \sqrt{\sum_j (x_j - z_j)^2}$$

Pairwise distances encode which observations are neighbors

Caveat: Feature Units

A large unit can dominate Euclidean distance



Raw magnitude can override the intended notion of similarity

Algorithm Families

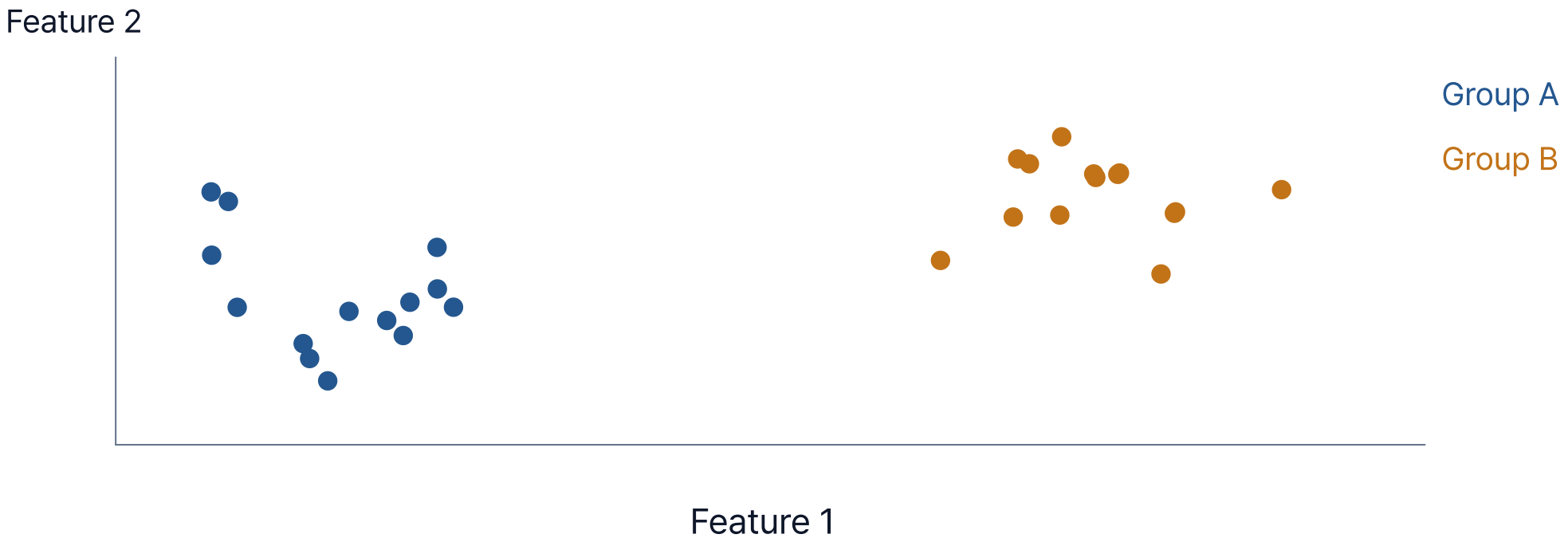
Choose the structure the model can represent

Family	Representation	Examples
Prototype	Centers	K-means, GMM
Density	Connected dense regions	DBSCAN, HDBSCAN
Hierarchy	Nested groups	Agglomerative
Graph	Neighbor connectivity	Spectral

Different families define a cluster differently

K-Means Objective

Assign points to a fixed number of centers



$$\min \sum_i \|x_i - \mu_{c_i}\|^2$$

Compact groups fit a center-based objective

Assignment Step

Each point chooses the nearest current center

Distance^2 A	Distance^2 B	Assignment
1	9	A
4	4	tie
9	1	B

A nearest-center rule partitions the feature space

Center Update

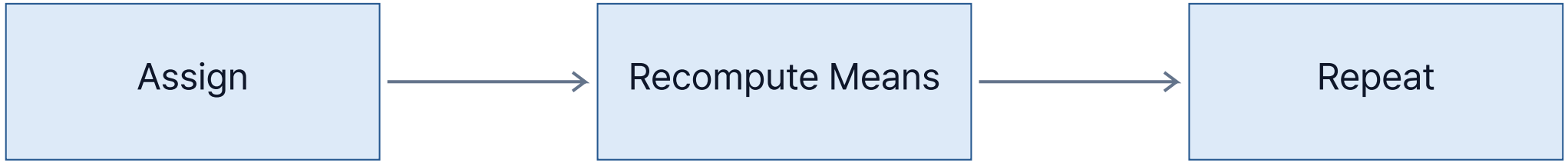
Each center becomes the mean of its assigned points

	x_1	x_2
Point 1	1	1
Point 2	2	2
Point 3	3	3
New Center	2	2

The new center is the coordinate-wise mean

Alternating Optimization

Repeat assignment and center updates



Each exact step does not increase the K-means objective

One K-Means Update

Assign six points and replace each center by its group mean

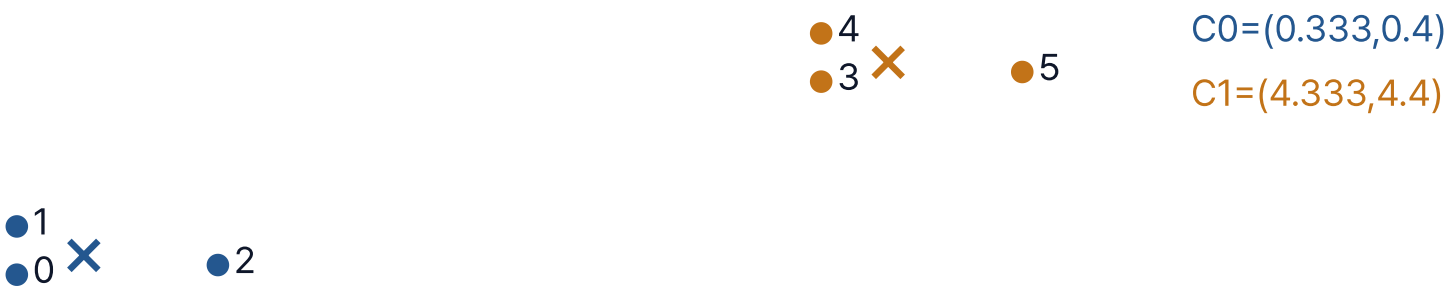
Point group after spread assignment	Updated center
(0,0), (0,1), (1,0.2)	(1/3, 0.4)
(4,4), (4,5), (5,4.2)	(13/3, 4.4)
Initial spread centers	(0,0), (5,4.2)

Demo: K-Means Iterations

Assignments and updated means are separate steps

Advance one complete assignment/update iteration Compare spread and nearby initial centers

Initial centers spread Complete iterations 1



Iteration 1; labels [0,0,0,1,1,1]; SSE 2.453333
Labels: last assignment; crosses: updated means

Answer: K-Means Iterations

The same input produces these calculated values

Input / quantity	Result
spread init; one update	centers (0.3333,0.4),(4.3333,4.4)
spread init; post-update SSE	2.453333
nearby init; first update	centers (0.5,0.1),(3.25,3.55); SSE24.5

Caveat: K-Means Iterations

A stable objective does not certify meaningful groups

Condition	Consequence
Empty cluster	Retain its previous center in this implementation
Different initial centers	May converge to different local optima

02 · Distributions and Density

01 Similarity and Prototypes

Choose a geometry

02 **Distributions and Density**

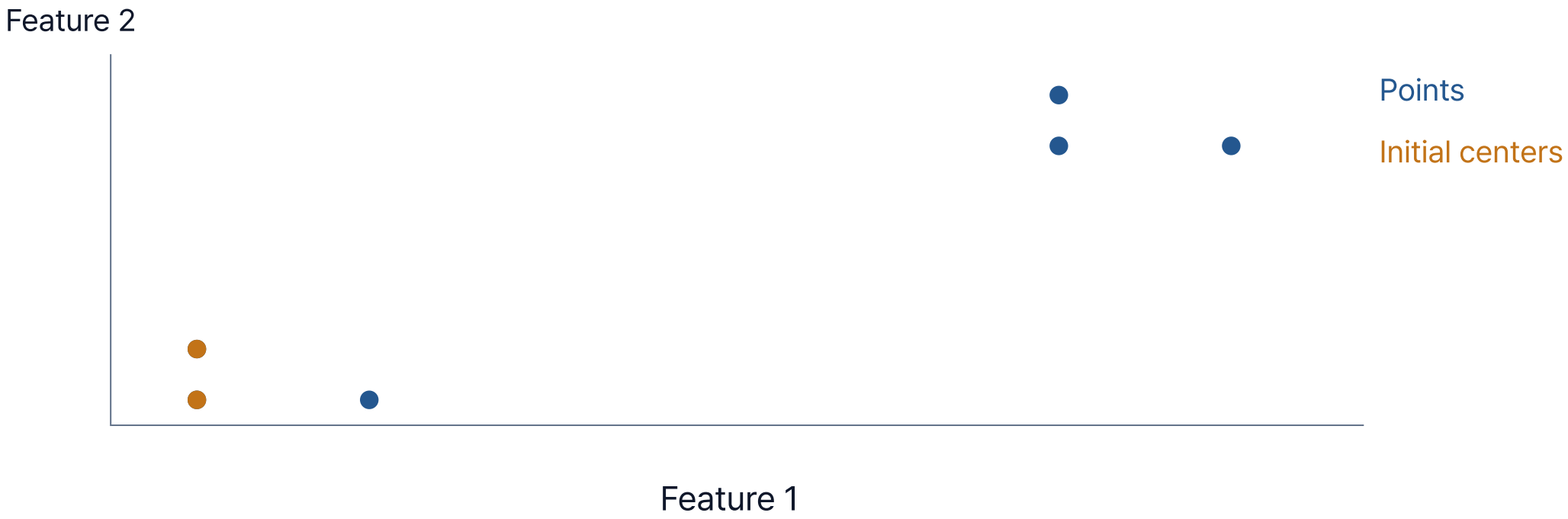
Represent flexible groups

03 Structure and Validation

Test the grouping

Initialization Sensitivity

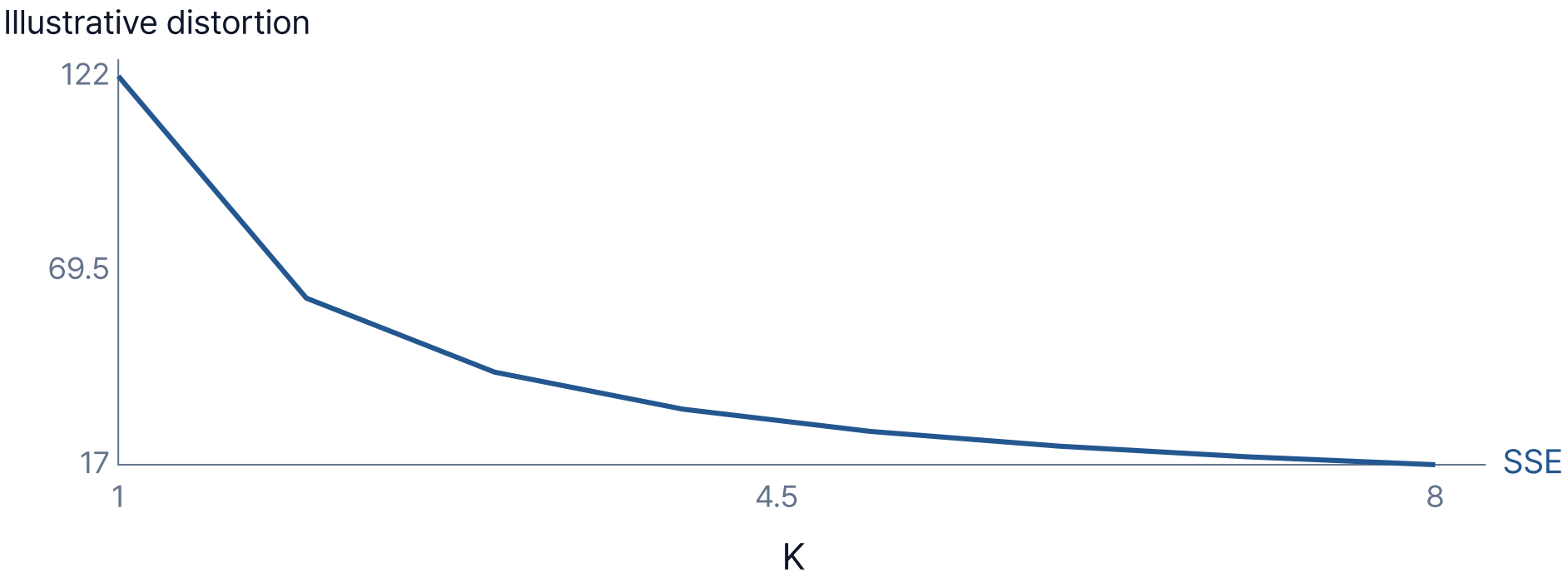
Different starts can lead to different local solutions



Both initial centers can start inside one dense region

Choosing the Cluster Count

Lower training distortion alone favors more centers



Adding centers keeps reducing this training objective

Silhouette Coefficient

Compare within-cluster and nearest-cluster distances

Within a	Nearest b	(b−a)/max(a,b)
2	5	0.6
4	3	-0.25

A negative value suggests a closer alternative cluster

Soft Assignments

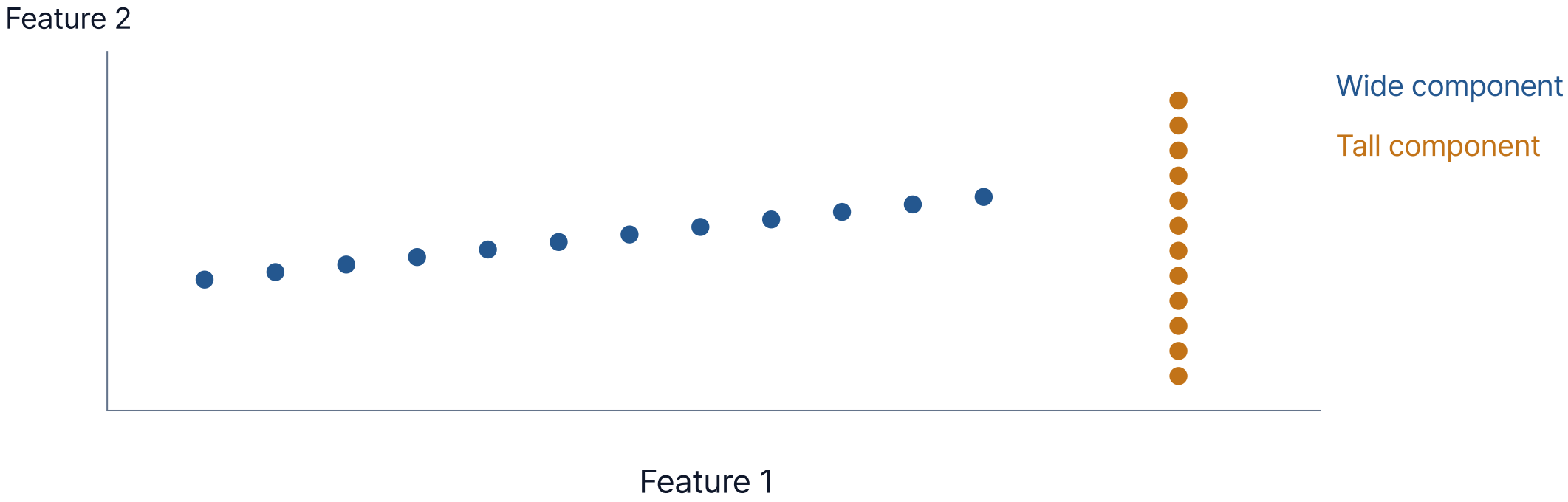
A mixture can assign fractional membership

	Component 1	Component 2
Point A	0.95	0.05
Point B	0.5	0.5
Point C	0.1	0.9

Boundary points need not have confident membership

Gaussian Mixture Model

Represent each component with a mean and covariance

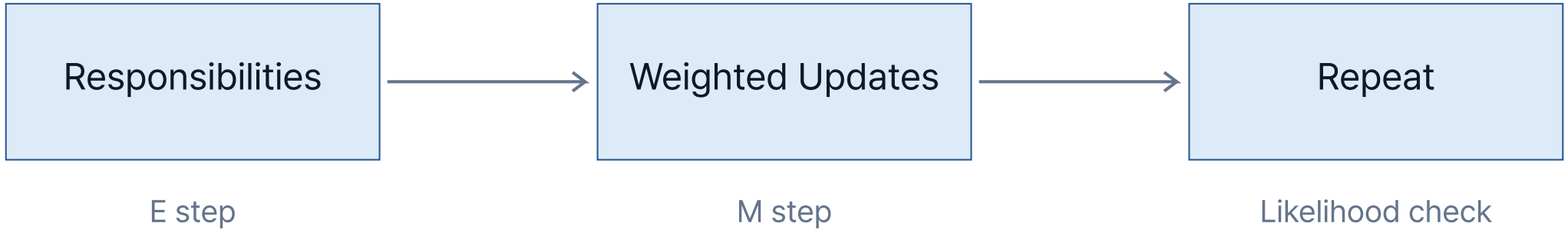


$$p(x) = \Sigma_k \ \pi_k \ N(x \mid \mu_k, \ \Sigma_k)$$

Covariance changes the shape of a component

Expectation-Maximization

Alternate membership estimates with distribution fitting



The weighted mean replaces the hard-assignment mean

Covariance Constraints

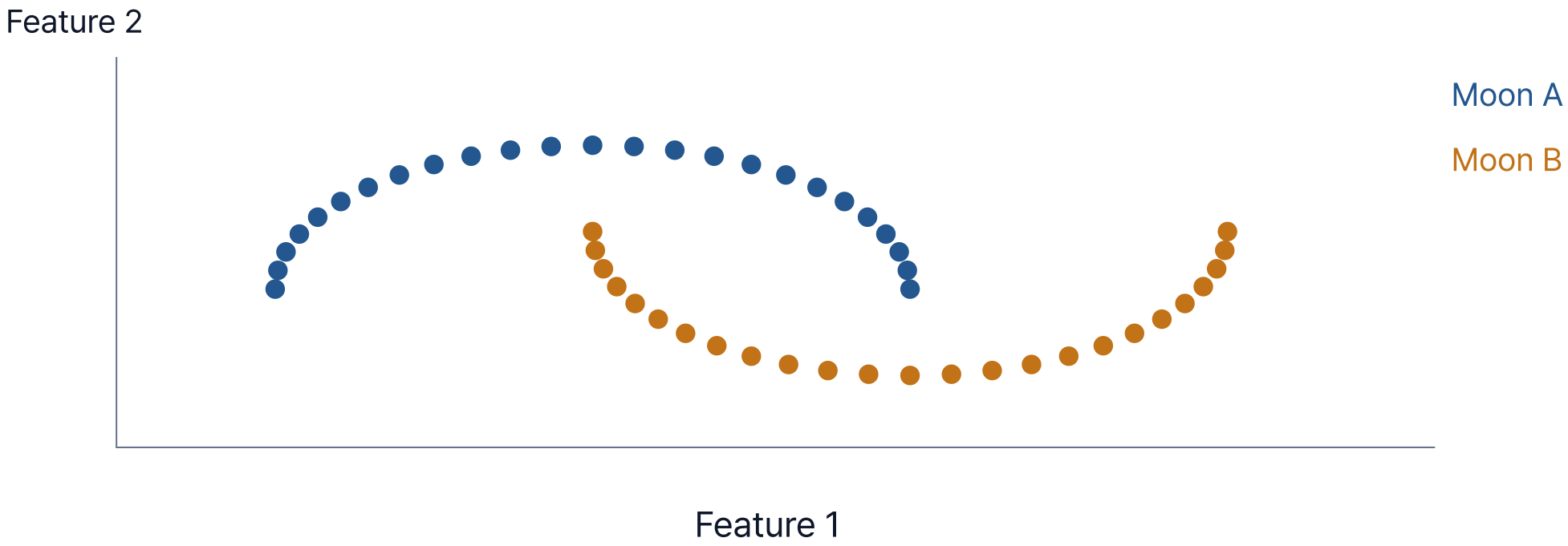
Restrict the shapes to match available evidence

Covariance	Geometry	Cost
Spherical	One radius	Few parameters
Diagonal	Axis-aligned ellipse	No correlation
Full	Rotated ellipse	More data needed

Flexible covariance needs more evidence per component

Caveat: Unequal Shapes

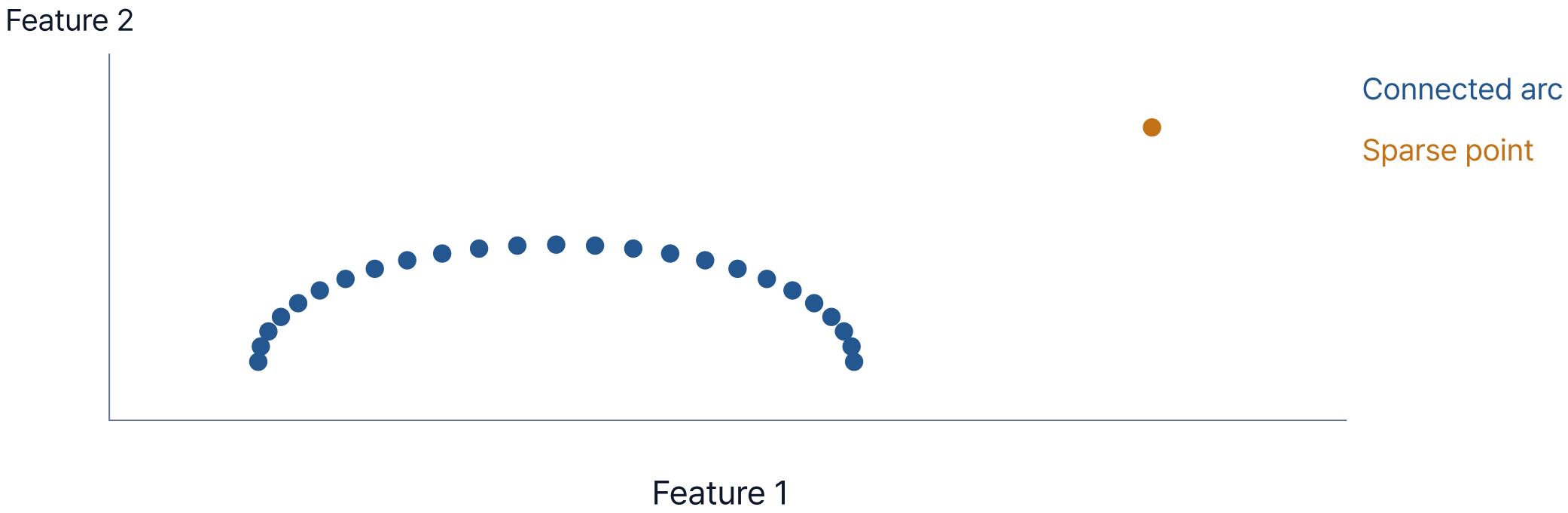
A center objective can cut through a curved group



Two curved groups challenge nearest-center geometry

Density Connectivity

Dense neighborhoods can follow a curved region



A sparse point can remain outside the dense component

Core, Border, and Noise

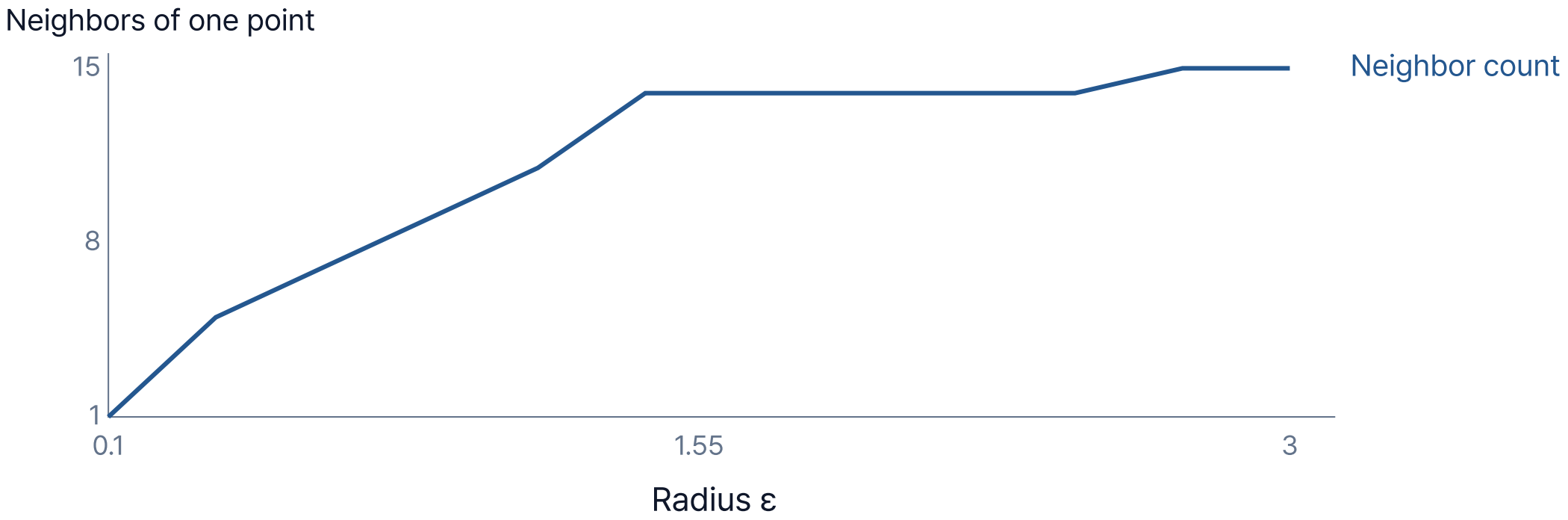
Neighborhood evidence determines each role

Role	Condition
Core	At least min_samples
Border	Near a core point
Noise	No density connection

A border point can join without being a core point

Density Parameters

A radius and minimum count set the resolution



Increasing the radius admits more observations

A Density Neighborhood

Core status depends on the epsilon neighborhood including self

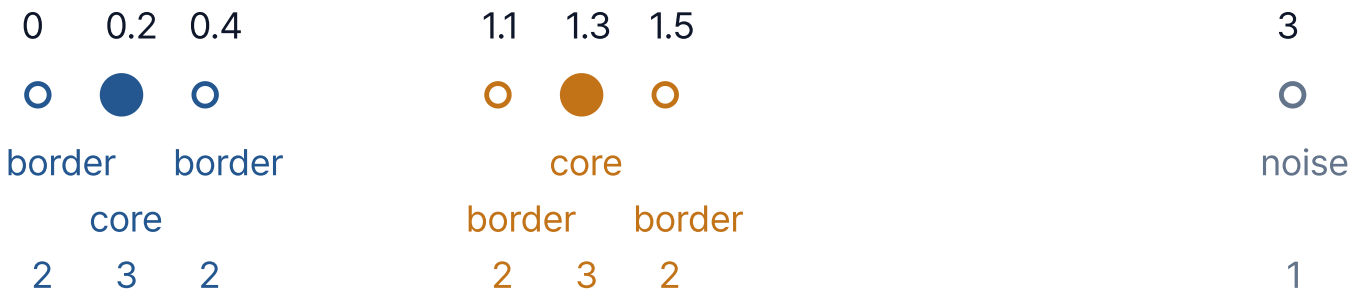
Input / setting	Calculation
x = 0, .2, .4, 1.1, 1.3, 1.5, 3	epsilon=.35; minPts=3
Neighbors of x=.2	0, .2, .4 → core
Neighbors of x=0	0, .2 → border
Neighbors of x=3	3 only → noise

Demo: Density Connectivity

Density connectivity can join groups through a chain

Change epsilon from 0.35 to 0.8 Count core points, clusters, and noise

Epsilon 0.35 MinPts (includes self) 3



Clusters 2; noise 1; core points 2

Bottom numbers count neighbors including self; labels [0,0,0,1,1,1,-1]

Answer: Density Connectivity

The same input produces these calculated values

Input / quantity	Result
x=[0,.2,.4,1.1,1.3,1.5,3]; minPts3	epsilon.35: 2 clusters, 1 noise
epsilon.8; minPts3	1 cluster, 1 noise
epsilon.35; minPts4	0 clusters, 7 noise

Caveat: Density Connectivity

One global density threshold can merge or erase structure

Condition	Consequence
epsilon too large	Core connectivity joins separate groups
minPts exceeds every neighborhood count	Every point is noise

03 · Structure and Validation

01 Similarity and Prototypes

Choose a geometry

02 Distributions and Density

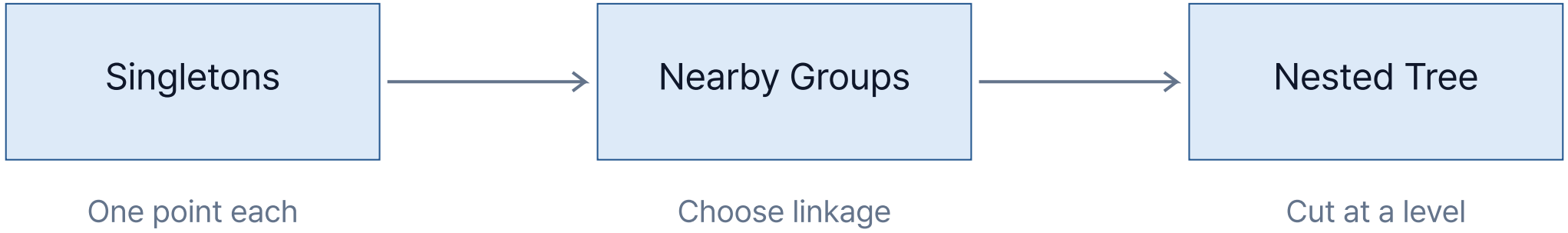
Represent flexible groups

03 **Structure and Validation**

Test the grouping

Hierarchical Clustering

Merge groups into a nested structure



The hierarchy retains several possible resolutions

Linkage Choices

Cluster distance depends on the chosen rule

Linkage	Merge Criterion
Single	Closest pair
Complete	Farthest pair
Average	Mean pairwise distance
Ward	Increase in within-group SSE

The same point distances can yield different merge sequences

Spectral Embeddings

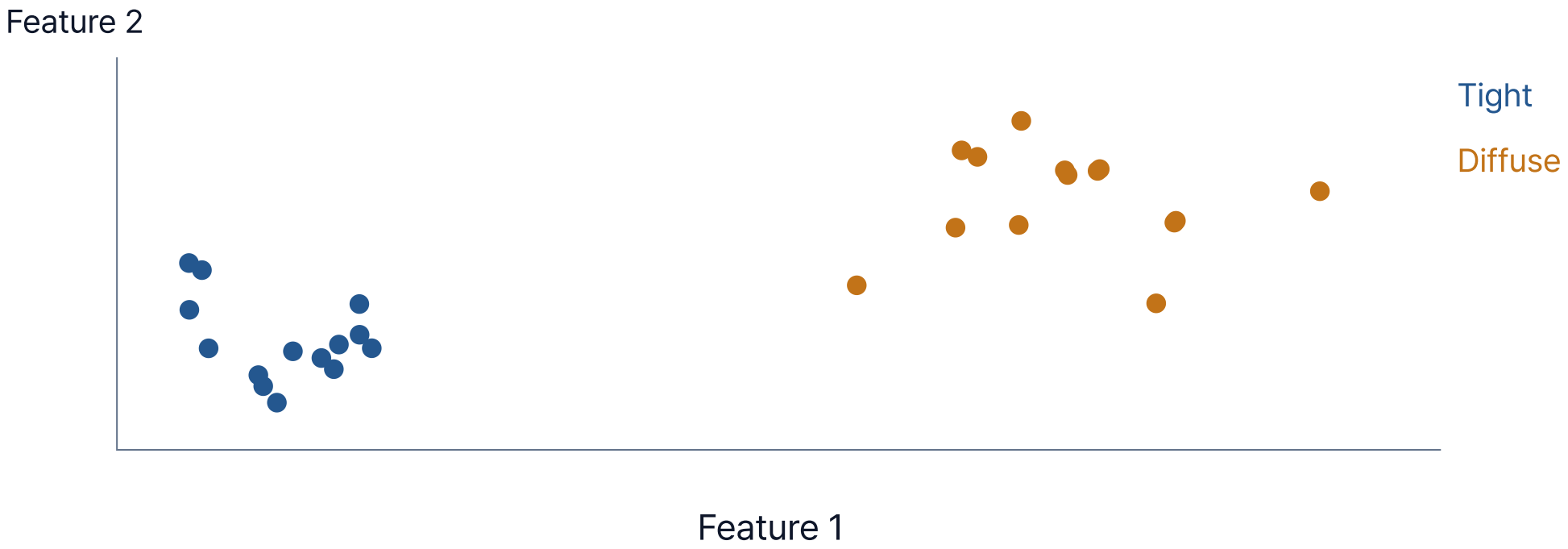
Use a neighbor graph before partitioning

	A	B	C	D
A	0	1	0	0
B	1	0	1	0
C	0	1	0	1
D	0	0	1	0

A graph records connectivity instead of only center distance

Variable-Density Groups

One global radius can fail across densities



One neighborhood radius faces different sampling densities

Internal and External Checks

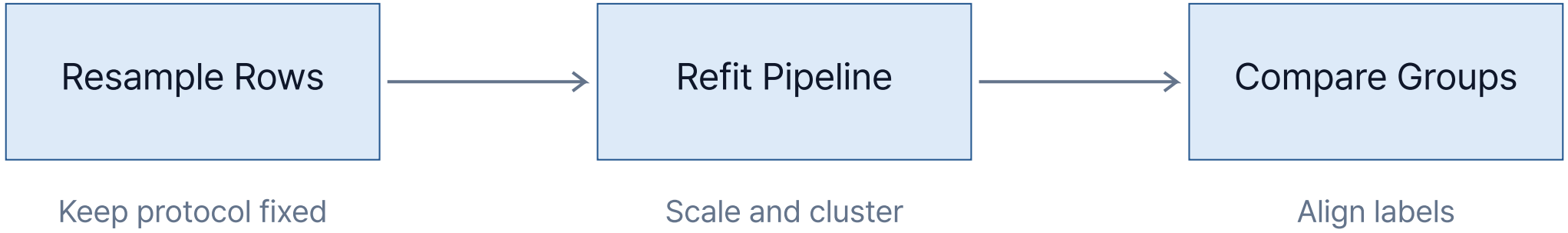
Evaluate structure separately from known labels

Check	Needs Labels?	Question
Silhouette	No	Distance separation
Adjusted Rand	Yes	Agreement with reference
Stability	No	Agreement after resampling
Profile	No	Useful feature differences

A grouping can be stable without matching a chosen reference

Stability Under Resampling

Repeat the full pipeline before trusting a grouping



Cluster identifiers can change even when memberships agree

Cluster Profiles

Translate assignments back into measured features

	Mean Load	Mean Vibration
Cluster A	0.2	1.8
Cluster B	1.7	0.3

Profiles explain which measurements distinguish the groups

A Minimal K-Means Loop

Read each assignment and update explicitly

```
X=np.array([[0.,0.],[0.,1.],[4.,4.],
[4.,5.]])
centers=X[[0,2]].copy()
for _ in range(5):
    labels=((X[:,None]-
centers)**2).sum(2).argmin(1)
    centers=np.array([X[labels==k].mean(0)
for k in range(2)])
print(centers)
```

```
[[0.  0.5]
 [4.  4.5]]
```

Caveat: Cluster Meaning

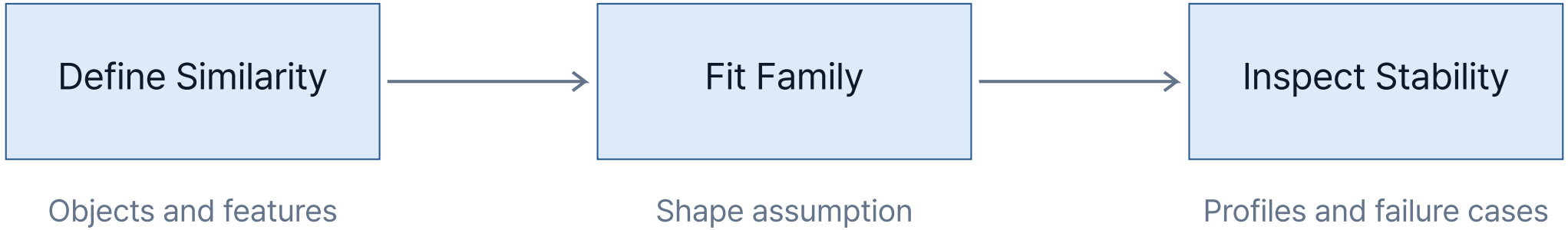
A mathematical partition does not establish a causal group

Choice	Consequence
Scaling	Changes neighbors
Chosen features	Changes similarity
Algorithm family	Changes group shape
Interpretation	Requires external evidence

Every grouping inherits the choices used to construct it

Selecting a Clustering Pipeline

Keep the representation and evaluation together



A useful cluster is a tested representation choice

Takeaway: Clustering

01 Define similarity before choosing an algorithm

02 Match the model to the group geometry

03 Inspect stability and measured profiles